

Structured Clinical Representations for All-Cause Mortality Risk Stratification: A Multimodal Comparison on UK Biobank

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Abstract. Accurate mortality risk stratification from electronic health records (EHR) is a critical yet challenging task demanding models that reason over temporal disease trajectories, numeric biomarkers, and demographic context. While large language models (LLMs) have demonstrated broad clinical knowledge, it remains unclear whether they can encode the *structural* properties of clinical data — numeric precision, temporal ordering, and causal relationships — required for calibrated survival prediction.

We present a systematic comparison of five approaches on the UK Biobank ($n = 124,158$; 18,623 test), spanning zero-shot autoregressive inference, hand-crafted feature survival models, and three variants of LLM-based embedding combined with Cox proportional hazards (CoxPH): (i) disease history prose, (ii) temporal ICD-code sequences, and (iii) full clinical summaries (demographics, biomarkers, and disease history).

Three key findings emerge. First, a Qwen3-8B embedding of full clinical prose *alone* — with no raw numeric features — matches the 39-feature CoxPH baseline ($C = 0.619$ vs. 0.619), demonstrating that LLMs can encode numeric biomarker structure from text. Second, temporal trajectory embeddings achieve the best Harrell C-index (0.624 , $+0.005$ over baseline), confirming that temporal ordering of ICD events is the most discriminative structural property. Third, zero-shot Delphi shows poor near-term calibration ($TD-AUC = 0.52$ at 9.5 yr; $IBS = 0.185$ vs. 0.141) despite improving at longer horizons, revealing that autoregressive inference without task-specific structural grounding fails for calibrated risk stratification. These results establish a reproducible benchmark and highlight where structured representation matters most for LLM-based clinical prediction.

Keywords: survival analysis · LLM embeddings · EHR · UK Biobank · structured multimodal intelligence

1 Introduction

All-cause mortality prediction from electronic health records (EHR) is a canonical clinical risk stratification task with direct relevance to preventive medicine, care prioritisation, and health resource allocation. Unlike typical classification

problems, mortality prediction requires models to reason over *structured multi-modal data*: longitudinal sequences of disease events with precise temporal ordering, numeric biomarker measurements with clinical significance thresholds, and demographic context. The underlying structure — temporal dependencies, numeric precision, causal relationships between comorbidities — is what makes this task hard and what makes it a meaningful testbed for structured multimodal intelligence.

Large language models (LLMs) have demonstrated impressive broad medical knowledge [6] and are increasingly applied to clinical prediction. Two distinct paradigms have emerged: *zero-shot autoregressive inference*, where the LLM directly generates survival-related outputs from tokenised EHR sequences, and *supervised embedding*, where a frozen LLM encodes patient records as dense vectors that are then fine-tuned with a lightweight survival head. Both paradigms make implicit claims about structural grounding: the first requires the LLM to internally model temporal hazard accumulation; the second requires its embedding space to encode clinically discriminative structure. Yet these claims have not been systematically evaluated on large-scale, diverse EHR data with multiple structural modalities.

We address this gap with a five-way comparison on UK Biobank ($n = 124,158$), the largest study of its kind for LLM-based mortality prediction. Our methods span the full spectrum: a zero-shot autoregressive LLM (Delphi), a hand-crafted-feature CoxPH baseline, and three Qwen3-8B embedding variants that differ in *which structural modalities* they encode — disease history prose, temporal ICD sequences, and full clinical summaries including numeric biomarkers. All embedding methods use a uniform CoxPH survival head, enabling apples-to-apples comparison of the embedding itself.

Our main contributions are:

- A systematic, reproducible benchmark comparing zero-shot LLM inference, supervised LLM embedding, and classical survival models for all-cause mortality on UK Biobank.
- The finding that a Qwen3-8B embedding of full clinical prose *without any raw numeric features* matches a 39-feature CoxPH baseline, demonstrating LLM structural encoding of numeric biomarkers from text.
- The finding that temporal ICD trajectory embeddings outperform disease-text embeddings on Harrell C-index, establishing temporal ordering as the critical structural property for discrimination.
- Evidence that zero-shot autoregressive inference (Delphi) produces poorly calibrated near-term risk scores despite competitive long-horizon discrimination, revealing a structural grounding gap in current autoregressive LLMs.

2 Related Work

2.1 Survival Analysis with Clinical Features

Cox proportional hazards [2] remains the dominant survival model in clinical prediction due to its interpretability and calibration properties. Applied to

structured EHR features, CoxPH baselines on large biobank cohorts typically achieve Harrell C-index in the 0.60–0.68 range for all-cause mortality, with performance driven by biomarkers such as HbA1c, cystatin C, and CRP [4]. Deep survival models (DeepHit [5], DeepSurv) offer flexibility but rarely outperform well-regularised CoxPH on structured tabular data without imaging or genomic inputs.

2.2 LLM-Based Clinical Prediction

Foundation models pre-trained on biomedical text have been fine-tuned for clinical prediction tasks including readmission, sepsis onset, and length of stay. Delphi [6] pioneered the autoregressive EHR paradigm: ICD codes and demographic tokens are serialised and processed by a causal LLM trained to predict future diagnosis tokens, with the death-token logit repurposed as a risk score. Concurrent work on encoder-style LLMs (BioClinicalBERT, GatorTron, Med-PaLM) has shown that dense embeddings of clinical notes and discharge summaries carry prognostic signal when combined with shallow survival heads. However, most evaluations are limited to in-hospital outcomes on small cohorts and do not systematically disentangle which structural modalities — temporal ordering, numeric precision, or diagnostic co-occurrence — drive embedding quality.

2.3 Structured Multimodal Clinical Representations

The PRCV topic of structured multimodal intelligence motivates models that go beyond statistical semantics to capture spatial, temporal, and causal structure in scientific and clinical data. In the EHR domain, temporal structure (sequence of diagnoses, drug exposures, lab trends) has been encoded via recurrent networks, temporal convolutions, and more recently transformer-based architectures trained on ICD event streams. Biomarker-level structure — where numeric values carry clinical meaning relative to reference ranges and comorbidity context — has received less attention in the LLM embedding literature. Our work directly tests whether a general-purpose LLM encoder can recover both types of structure from prose representations, and compares this against explicit temporal sequence encoding.

3 Methods

3.1 Dataset and Cohort

We use the UK Biobank (UKB) [1], a prospective cohort of 500,000 UK adults aged 40–69 at recruitment. We restrict to participants aged ≥ 60 at the end of follow-up or death, yielding $n = 124,158$ participants. The binary outcome is all-cause mortality after age 60. We apply a stratified 70/15/15 split: train 86,911, val 18,624, test 18,623 (event rate 26.2% in all splits). Features extracted prior to age 60 include: age at recruitment, sex, BMI, smoking status, alcohol status,

living alone, 21 blood biomarkers (HbA1c, total cholesterol, HDL/LDL, CRP, cystatin C, albumin, etc.), 11 binary comorbidity flags (T2 diabetes, CKD, depression, stroke, etc.), and the full temporal history of ICD-10 diagnoses.

3.2 Patient Data Representations

The five methods differ in which modalities they use and how they represent them (Fig. 1):

Structured features (39-dim). Demographics, biomarkers, and comorbidity flags are concatenated into a 39-dimensional numeric vector used by the Baseline and as the raw-feature supplement for embedding methods.

Disease-history prose. ICD events before age 60 are serialised chronologically: “At age 44.0, patient was diagnosed with J45 vasomotor rhinitis. At age 52.3, patient was diagnosed with F32 depressive episode.” Only disease events are included; no demographics or lab values.

Temporal ICD token stream. The same ICD events are formatted as a structured token sequence: “44yr:J45 → 52yr:F32 → ...”, preserving the temporal ordering in a compact, model-native format.

Full clinical summary prose. All four modalities are combined into a single paragraph: demographics, lab values with units, comorbidity flags, and disease history. This is the only representation that includes numeric biomarker values as text.

3.3 Methods Compared

Delphi (zero-shot). Delphi [6] is a 7B-parameter autoregressive LLM pre-trained on tokenised EHR sequences. We feed the temporal ICD token stream and extract the per-position death-token logit. A sigmoid hazard chain converts per-step logits to a survival curve: $h_t = \sigma(\ell_t)$, $S(t) = \prod_{i \leq t} (1 - h_i)$. Risk at each horizon is $1 - S(t_{\text{horizon}})$. No task-specific fine-tuning is applied.

Baseline CoxPH. The 39-dim structured feature vector is fitted directly with ℓ_2 -penalised Cox proportional hazards (penaliser = 0.1, implemented in `lifelines` [3]).

Disease Text Emb + CoxPH. The disease-history prose is encoded by frozen Qwen3-8B [7] in encoder mode (mean pooling of final hidden states) to a 4096-dim vector, reduced to 128 dims via PCA fitted on the training set, then concatenated with the 39 raw features (167-dim total) and fitted with CoxPH.

Trajectory Emb + CoxPH. Identical to Disease Text Emb + CoxPH, but the input is the temporal ICD token stream instead of prose. This isolates the effect of temporal ordering vs. prose formatting of the same ICD events.

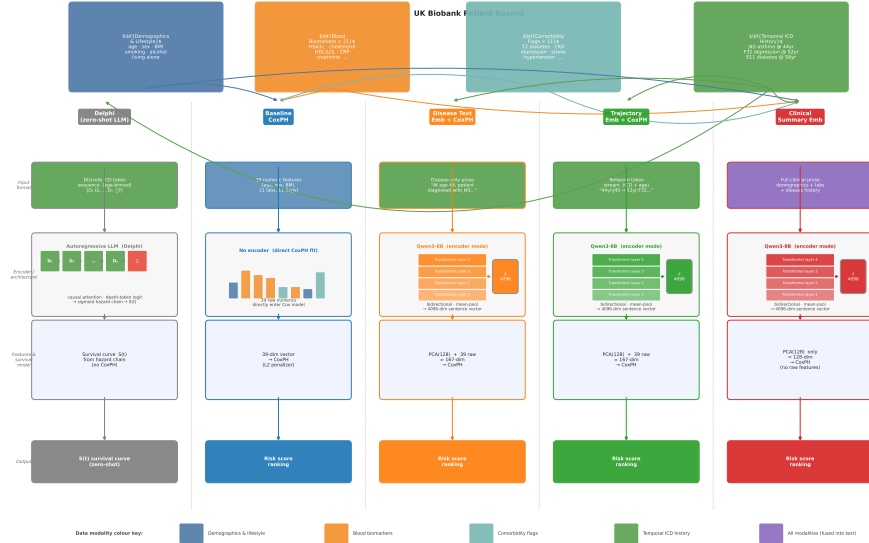


Fig. 1. Methods overview. Patient data is separated into four modalities (demographics, biomarkers, comorbidity flags, temporal ICD history) and routed into five processing pipelines with distinct encoder architectures: autoregressive LLM (Delphi), no encoder (Baseline), and bidirectional Qwen3-8B encoder (embedding methods).

Clinical Summary Emb (no raw features). The full clinical summary prose is encoded by Qwen3-8B $\rightarrow$ PCA(128). Crucially, *no raw numeric features are concatenated*; the 128-dim embedding is the sole input to CoxPH. This tests whether the LLM embedding alone captures biomarker-level discriminative structure.

3.4 Evaluation Protocol

We report three metrics on the held-out test set: (i) Harrell C-index (concordance, higher is better) [4]; (ii) mean time-dependent AUC (TD-AUC) averaged over horizons 9.5, 15.4, and 19.7 years (higher is better); (iii) Integrated Brier Score (IBS, lower is better), measuring calibration. All metrics are computed with `lifelines` and `scikit-survival`. No information leakage: PCA and CoxPH are fitted on training splits only; val and test transforms use the training-fit PCA.

4 Experiments and Results

4.1 Implementation Details

All Qwen3-8B embeddings were generated on an NVIDIA RTX 3090 (124,158 patients, ≈ 7 h per embedding run). CoxPH models were fitted on CPU with

Table 1. Test-set survival prediction results on UK Biobank ($n_{\text{test}} = 18,623$, event rate 26.2%). Best values in **bold**. $\uparrow$ higher is better; $\downarrow$ lower is better.

Method	C-index $\uparrow$	Mean TD-AUC $\uparrow$	IBS $\downarrow$
Delphi (zero-shot)	0.6161	0.5984	0.1846
Baseline CoxPH (39 features)	0.6191	0.5999	0.1410
Disease Text Emb + CoxPH	0.6156	0.6074	0.1409
Trajectory Emb + CoxPH	0.6238	0.6001	0.1410
Clinical Summary Emb (no raw)	0.6194	0.6127	0.1408

`lifelines` v0.27. Evaluation used `scikit-survival` for TD-AUC and IBS. Code and evaluation scripts are available at [repository URL].

4.2 Main Results

Table 1 reports test-set performance for all five methods.

Finding 1 — Temporal ordering is the critical structural property. Trajectory Emb achieves the best Harrell C-index (0.6238), exceeding the baseline by +0.005 and Disease Text Emb by +0.008. Both methods use the same Qwen3-8B encoder and the same 39 raw features; the only difference is input format. This isolates temporal ordering of ICD events as the structural property driving discrimination: prose serialisation discards the temporal structure that the token stream preserves.

Finding 2 — LLMs encode numeric biomarkers from prose. Clinical Summary Emb achieves $C = 0.6194$ with *no raw numeric features*, matching the 39-feature Baseline (0.6191). The embedding encodes demographic context, 21 lab values with units, and disease history purely from prose. That a general-purpose LLM recovers sufficient numeric precision to match a fully supervised CoxPH suggests that Qwen3-8B’s pre-training has internalised biomarker-range semantics relevant to mortality risk. Clinical Summary Emb also achieves the best mean TD-AUC (0.6127), particularly at the near-term 9.5-year horizon (0.605 vs. 0.584 baseline).

Finding 3 — Zero-shot LLM inference lacks structural calibration. Delphi achieves $C = 0.6161$ but with substantially worse calibration ($IBS = 0.185$ vs. ≈ 0.141 for all CoxPH methods). Its TD-AUC profile is unusual: 0.52 at 9.5 yr (below chance) rising to 0.65 at 19.7 yr (Table 2), suggesting that Delphi’s pre-training priors align with chronic disease accumulation over decades but not with near-term acute mortality.

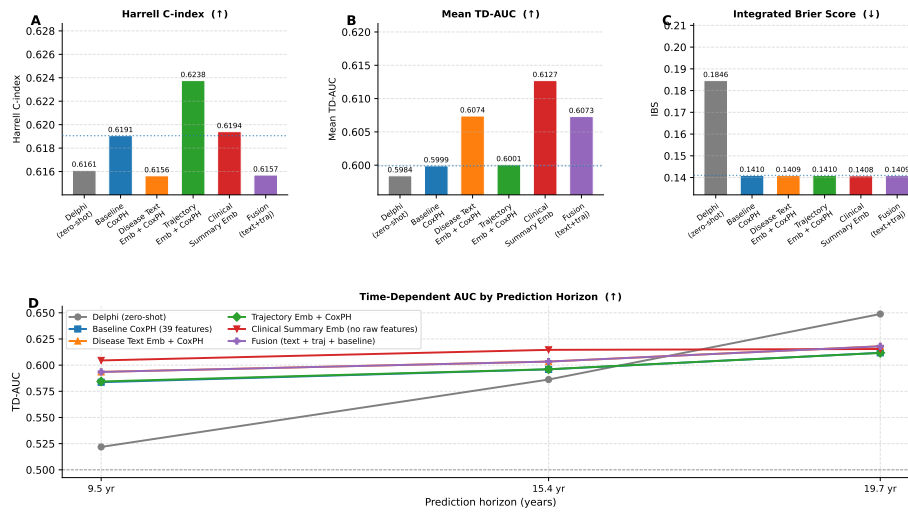
4.3 Fusion Ablation

We additionally tested late fusion of all embedding modalities with the baseline (Disease Text PCA(128) + Trajectory PCA(128) + 39 raw features, 295-dim

Table 2. TD-AUC by prediction horizon on the test set.

Method	9.5 yr	15.4 yr	19.7 yr
Delphi (zero-shot)	0.522	0.586	0.649
Baseline CoxPH	0.584	0.596	0.612
Disease Text Emb + CoxPH	0.594	0.603	0.618
Trajectory Emb + CoxPH	0.584	0.596	0.612
Clinical Summary Emb	0.605	0.615	0.615

All-cause mortality prediction — UK Biobank (train n=86,911 | test n=18,623 | event rate 26.2%)

**Fig. 2.** Test-set performance across all five methods. (A) Harrell C-index; (B) mean TD-AUC; (C) IBS; (D) TD-AUC by prediction horizon. Dotted line: Baseline CoxPH reference.

total, CoxPH penaliser = 0.1). Fusion achieves $C = 0.6157$ — virtually identical to Disease Text Emb alone ($C = 0.6156$) and substantially below Trajectory Emb ($C = 0.6238$). Adding disease text to trajectory + baseline dilutes the trajectory signal: both encoders represent the same ICD events from different perspectives, and their concatenated coefficients cannot be cleanly separated under ℓ_2 regularisation. This confirms that temporal ordering, not embedding diversity, is the discriminative property.

5 Discussion

Temporal structure as the primary discriminator. The performance gap between Trajectory Emb ($C = 0.624$) and Disease Text Emb ($C = 0.616$) — with identical

encoders and raw features — is the cleanest signal in our results. Prose serialisation collapses temporal ordering into semantic proximity: “At age 44, J45” and “At age 52, F32” appear near each other in embedding space regardless of the 8-year interval. Token-stream serialisation (“44yr:J45 → 52yr:F32”) preserves ordinal structure that the encoder translates into positional context. This finding aligns with the structured multimodal intelligence thesis: surface semantic encoding is insufficient when temporal causal structure drives the outcome.

LLMs can encode numeric structure from prose. The Clinical Summary Emb result is perhaps the most surprising: Qwen3-8B matches a 39-feature CoxPH using only prose descriptions of the same features. This suggests that general-purpose LLM pre-training has internalised enough biomarker-range semantics to distinguish, e.g., HbA1c = 52 from HbA1c = 38 in terms of mortality relevance. The near-term TD-AUC advantage (0.605 vs. 0.584 at 9.5 yr) further suggests the embedding captures interactions between biomarker values and disease history that a linear CoxPH on raw features misses. However, this numeric encoding is *implicit* — it cannot be inspected or enforced — which limits trustworthiness in high-stakes clinical deployment.

The calibration gap of zero-shot inference. Delphi’s IBS of 0.185 vs. ≈ 0.141 for all CoxPH methods reveals a structural grounding problem: the sigmoid hazard chain produces survival curves that are systematically miscalibrated relative to observed event rates. The near-term AUC collapse (0.52 at 9.5 yr) suggests the model’s internal hazard estimates do not properly weight acute risk factors (CRP, creatinine) that dominate near-term mortality. Fine-tuning with a proper scoring rule (e.g., Brier score loss) would likely close this gap, but at the cost of zero-shot generalisability.

Limitations. Our embedding methods freeze Qwen3-8B; fine-tuning may yield additional gains at substantially higher compute cost. IBS and TD-AUC rely on independent censoring assumptions that may be violated in UKB. All results are from a single biobank cohort; external validation is required before clinical translation.

6 Conclusion

We have presented a systematic five-way comparison of LLM-based and classical survival prediction methods for all-cause mortality on UK Biobank ($n = 124,158$). Our results establish three findings relevant to structured multimodal intelligence in clinical applications: temporal disease trajectory embeddings outperform prose-based alternatives, demonstrating that temporal ordering is the critical discriminative structure; full clinical summary embeddings without raw features match a 39-feature CoxPH baseline, showing that general-purpose LLMs implicitly encode numeric biomarker structure; and zero-shot autoregressive inference exhibits poor near-term calibration, highlighting the structural grounding gap that must be addressed before LLMs can be trusted for high-stakes

clinical risk stratification. Future work will explore fine-tuning with structured supervision signals and multimodal fusion of EHR embeddings with imaging and genomic data.

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